Constraining the Apsidal Constant of Spica by Modelling Ellipsoidal Variations

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Spica is a massive binary star system that shows apsidal motion

Apsidal Motion

results from the rotationally and tidally distorted shapes of the two stars

these distortions lead to ellipsoidal variations in the light curve

is the advancement of the longitude of periastron

periastron is the point in an elliptical orbit where the two stars are closest

depends in part on apsidal period and eccentricity

which together determine the apsidal constant k_2, a measure of the primary star's mass distribution

The model light curve

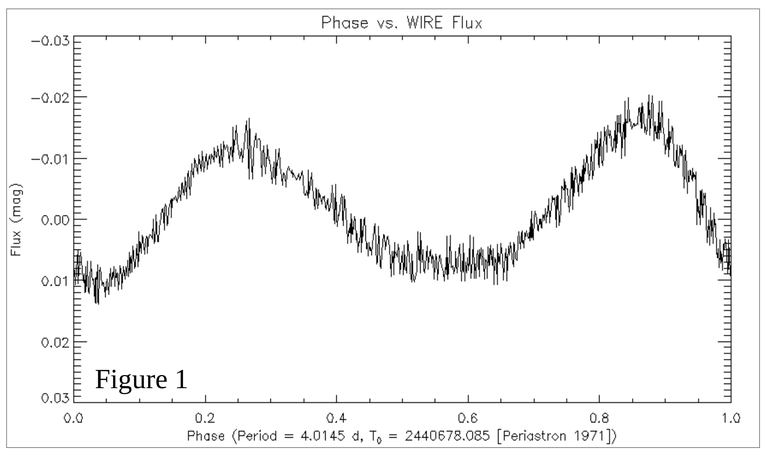
shows ~ 4-day ellipsoidal variations, which constrain the orbital eccentricity (e) and apsidal period (U)

fits best for e = 0.065 ± 0.015 and U = 105 ± 2 years, yielding $k_2 = -2.52^{+0.14}_{-0.21}$ consistent with $k_2 = -2.4$ (Ref. 3)

requires a high number (500) of longitude points over the stellar surface for numerical accuracy

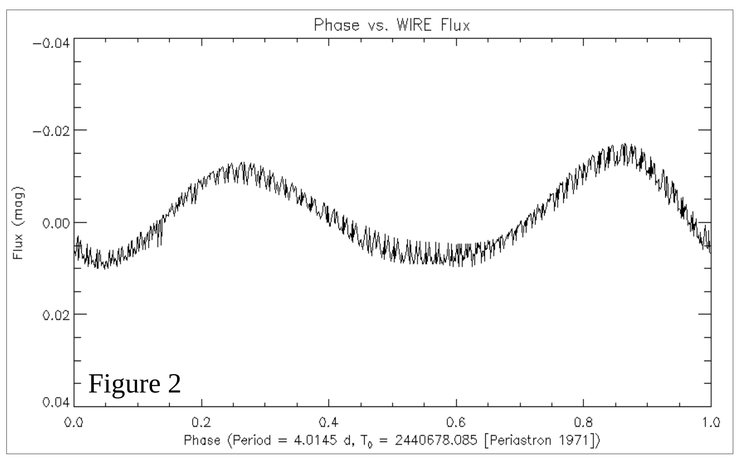
lacks the reflection effect, which could improve the model fit at phase = 0.56

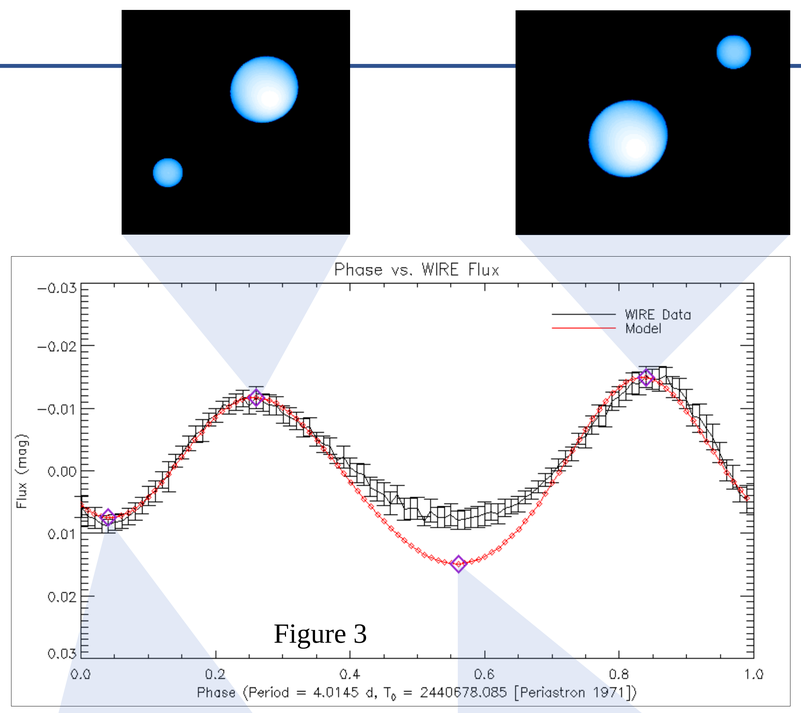
We modeled the WIRE light curve using a numerical model to constrain values for the apsidal period and eccentricity, which yield an apsidal constant consistent with theory



Filtering Out the High Frequency Pulsations to Isolate Ellipsoidal Variations

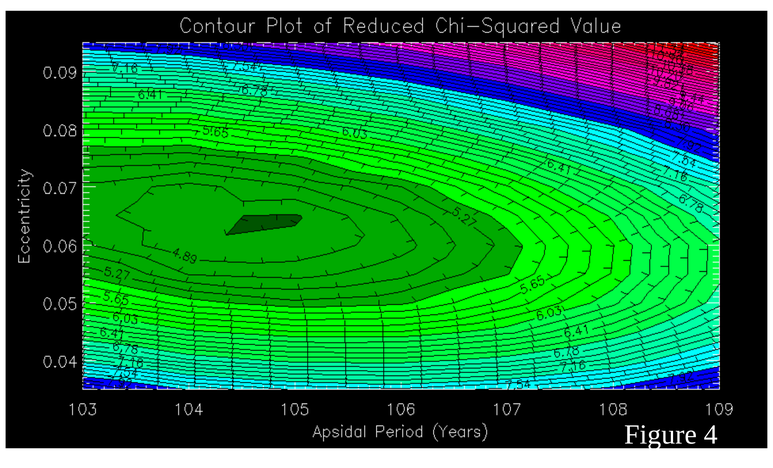
Photometric data from the Wide Field Infrared Explorer (WIRE) gives the light curve of Spica over 24 days in 2005 (Ref. 2). Figure 1 shows Spica's light curve from the WIRE data phased over its four-day orbital period. The change in relative magnitude, due to the changing perspectives of the system's distorted primary star, has an amplitude of about ± 1%. Since the primary is also a pulsating variable, the light curve has higher-frequency fluctuations with smaller amplitudes, ~ 0.5%, and a period of about four hours. We are only concerned with the orbital motion of Spica so these pulsations only contribute as "noise" which we wish to be filtered out. The Digital Filter function in Interactive Data Language (IDL) was used to set a bandpass around Spica's orbital frequency, $(4.0145 \text{ days})^{-1} = 0.249 \text{ d}^{-1}$, to remove anything but the variations from the orbital motion. This IDL function also decreases the amplitude by $2\cdot(\text{bandwidth})^{-1}$ and since the bandpass was set between 0.248 and 0.250 d^{-1}, the light curve had to be renormalized by multiplying by 1000. In addition, the filtering had to be done in the time domain, rather than the phase domain, to work properly so the original WIRE data was filtered and then phased over the orbital period, which can be seen in Figure 2. With data from multiple orbital periods phased over one, errors bars for the magnitude can be estimated from the averaged light curve. Figure 3 shows the averaged phased light curve with error bars binned over 100 phase points together with renderings of the binary at the maxima and minima.

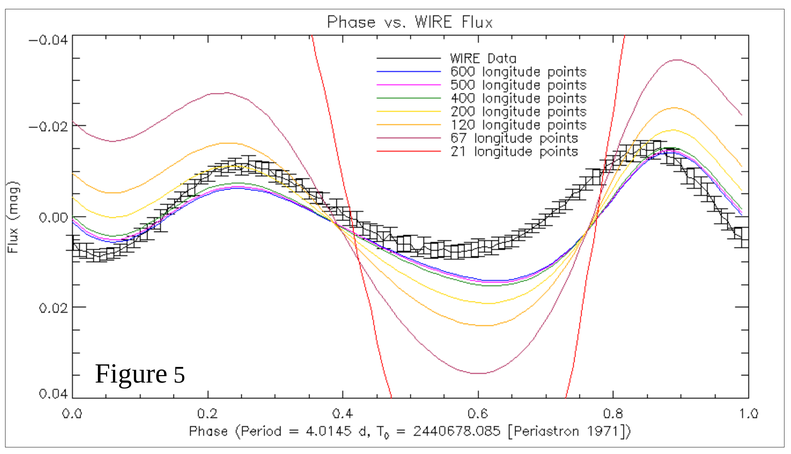




Model Light Curve Most Sensitive to Eccentricity & Apsidal Period

The numerical model for Spica (Ref. 1, 4-8) depends on 15 parameters that have to be set for each run. Six of these parameters were adjusted in turn to test for sensitivity to the model light curve: inclination, apsidal period, eccentricity, mass ratio, semi-major axis and the longitude of periastron. The light curve is most sensitive to the eccentricity and the apsidal period. The values for both were found via grid search and are 0.065 ± 0.015 and 105 ± 2 years for the eccentricity and the apsidal period, respectively. Figure 4 shows the contour plot of the reduced chi-squared surface (lowest value 4.65 with 98 degrees of freedom) between the model light curves and the observed light curve in Figure 5.





High Spatial Resolution of Stellar Surfaces Required for Numerical Accuracy

We find inadequate spatial sampling of the stellar surfaces leads to model light curves which grossly over predict the amplitude of ellipsoidal variations relative to observations. While the number of longitude points has a great impact on the amplitude of the light curve, the number of latitude points makes little difference. Figure 5 shows seven model light curves at different spatial resolutions in longitude plotted over the phased light curve data from WIRE. The 21-longitude-point model can be seen as the red light curve running off the plot and shows how drastically the spatial resolution can change the light curve. The model light curves show less sensitivity to longitude resolution around 400 points and between 500 and 600 points there is very little difference. Polychromatic light curve models with 210 wavelength points, 67 latitude points, 500 longitude points, and 100 phase points take 4 hours and 40 minutes to run. Running the models monochromatically, at 5000 Å, has no significant impact on the shape of the light curve and significantly reduces the runtime to 10 minutes. With this runtime it was feasible to run models for parameter estimation.

Reflection Effect to be Tested

There is a reflection effect between the primary and secondary where one star heats up the other through an exchange of photons. While the model is able to account for it, the time required with the current code and hardware is impractical. Currently, to compute synthetic photometry for a single phase point, out of 100, with the reflection effect took 127 hours with 210 wavelength points. However, comparing the flux from this computation to the flux of the same orbital phase (0.56) in the best fit model in Figure 3 suggests that this effect could account for the discrepancy between the model and the WIRE data. The next steps in this project would be to find a way to run the models with reflection in a time efficient manner and test how much of an impact it will have on the whole light curve. In addition to that, model sensitivity to wavelength resolution could also be tested further.

References

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